DS 223 Marketing Analytics

Network Science in Marketing

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Community Detection

Objective

- Girvan–Newman algorithm
- Infomap

Communities

Network communities are groups of nodes such that vertices inside the group connected with many more edges than between groups.

What makes a community?

What makes a community (cohesive subgroup):

- **Mutuality of ties:** almost everyone in the group has ties (edges) to one another
- **Compactness:** closeness or reachability of group members in small number of steps, not necessarily adjacency
- **Density of edges:** high frequency of ties within the group
- **Separation:** higher frequency of ties among group members compared to non-members

Idea

Graph Density:

$$\rho = \frac{m}{n(n-1)/2}$$

Community Internal Density:

$$\delta_{int} = \frac{m_s}{n_s(n_s-1)/2}$$

External Density:

$$\delta_{ext} = \frac{m}{n(n-1)}$$

$$\delta_{int} > \rho, \ \delta_{ext} < \rho$$

Modularity Score

Apart from the density score, the effectiveness of community detection can be assessed by Modularity Score. The edges within the cluster to **expected** in random graph with identical degree sequence.

$$Q = \sum_{u} \left(\frac{m_u}{m} - \left(\frac{k_u}{2m} \right)^2 \right)$$

where:

- k_u : internal nodes
- $\frac{m_u}{m}$: the ratio of internal edges to total edges

$$Q \in [-0.5, 1)$$

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Note, when $Q = 0$, then we are dealing with single community.